Demand Estimation, Markups, and Merger Simulation

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Abstract

Using a simulated differentiated-products dataset of 247 products across 10 regional markets and 11 firms, this paper estimates demand, recovers markups, and simulates a horizontal merger. Demand is estimated via the Berry (1994) logit inversion, with two cost shifters instrumenting for price; the IV price coefficient is -2.55, slightly smaller in magnitude than OLS, consistent with positive price–quality correlation. Multi-product Bertrand–Nash first-order conditions recover a median Lerner index of 0.12, and pure logit imposes the well-known restriction that a firm sets a common markup across its within-market products. A counterfactual merger between two firms raises merging-product prices by roughly 1.0 percent on average while leaving rival prices essentially unchanged, the textbook unilateral-effects signature, with per-consumer compensating-variation losses concentrated in the few markets where the merging firms hold meaningful share. The exercise closes by mapping the methodology onto a hospital-merger application and critiquing its limits when prices are negotiated rather than posted.

Keywords: demand estimation; multinomial logit; Berry inversion; Bertrand–Nash markups; merger simulation; unilateral effects.
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Plain-Language Summary

Antitrust regulators face a hard question before a merger happens: will it raise prices? This paper works through the standard economic toolkit for answering that question, using a simulated dataset of 247 products sold by 11 firms across 10 markets.

The method has three steps. First, estimate how sensitive buyers are to price, carefully separating a product's price from its unmeasured quality, which would otherwise distort the estimate. Second, back out each firm's markup, the gap between price and cost, from how it competes. Third, simulate what happens when two firms combine into one.

Merging the two firms raises their own prices by about 1 percent and barely touches competitors, the classic pattern when a merger hands a firm a little more pricing power. The harm to consumers is small here, and largest in the handful of markets where the two firms together actually have meaningful presence.

The cleanest caveat is about transferability. The model assumes firms post prices and buyers react. The paper closes by showing that in a real setting such as hospital mergers, where prices are negotiated between hospitals and insurers, this exact machinery gives precise but wrong answers, a reminder that a tractable model applied outside its domain can mislead.

OVERVIEW

The dataset records $T = 10$ regional markets, 11 national firms, and $J = 247$ unique products. Each product is offered in exactly one market. For each product the data report the market identifier, the firm, four characteristics (a constant plus three product attributes), the price, two cost shifters, and the market share. The exercise estimates demand under two specifications, recovers Bertrand-Nash markups, simulates a counterfactual merger between firms 1 and 10, and reflects on applying the methodology in practice. All estimation is performed in R; results are reproducible via the master script `Dalis_Abdallah_main.R`.

1 MARKET STRUCTURE

Table 1 reports, for each market, the number of active firms, the number of products on offer, and the inside-good share, the fraction of agents who purchased any inside good rather than the outside option. The number of firms ranges from 7 to 11 across markets, and product counts range from 11 (market 10) to 49 (market 3). Inside-good shares range from 0.418 in market 9 to 0.828 in market 7, indicating substantial variation in the strength of the outside option across markets. Markets 3 and 7 are the most "captured" (small outside option), while markets 9 and 10 leave more than half of agents to the outside good.

2 PRODUCT-LEVEL SUMMARY STATISTICS

Table 2 reports the mean, median, minimum, maximum, and standard deviation of each variable across all $J = 247$ products. The first column ch_1 is a constant of 1 for every product (the

Table 1: Market-level summary: active firms, products offered, and inside-good share.

Market	Firms active	Products	Inside share
1	11	30	0.60
2	8	24	0.56
3	11	49	0.78
4	7	15	0.53
5	9	21	0.61
6	7	13	0.50
7	11	43	0.83
8	10	24	0.75
9	8	17	0.42
10	8	11	0.42

model's intercept), so it is omitted. Shares are right-skewed: mean 0.024 but median 0.010, with a maximum of 0.27. Prices have a median of 3.52 and span $[1.30, 6.21]$. The two cost shifters look broadly uniform on $[0, 0.5]$, which is consistent with how a clean simulation typically generates exogenous instruments.

Table 2: Product-level summary statistics ($n = 247$). Characteristics ch_2, ch_3, ch_4 are the non-constant product attributes; $ch_1 = 1$ is the intercept.

Variable	Mean	Median	Min	Max	SD
Share	0.024	0.010	0.000	0.269	0.038
ch_2	1.474	1.425	-1.166	4.363	0.967
ch_3	1.498	1.452	-1.147	3.936	0.913
ch_4	1.472	1.483	-1.388	4.483	0.986
Price	3.502	3.520	1.297	6.213	0.829
Cost shifter 1	0.264	0.272	0.006	0.500	0.141
Cost shifter 2	0.239	0.237	0.002	0.500	0.140

3 PURE MULTINOMIAL LOGIT

Under a pure multinomial logit, consumer i in market m chooses among J_m inside goods plus an outside option $j = 0$. The indirect utility from product j is

$$u_{ij} = x_j^\top \beta - \alpha\, p_j + \varepsilon_{ij}, \qquad u_{i0} = \varepsilon_{i0}, \tag{1}$$

where x_j is the vector of observed characteristics, p_j is price, and ε_{ij} is i.i.d. type-I extreme value across consumers and products. The outside-good utility is normalised to its idiosyncratic shock alone. Under these assumptions, market shares take the closed form

$$s_j = \frac{\exp(x_j^\top \beta - \alpha p_j)}{1 + \sum_{k=1}^{J_m} \exp(x_k^\top \beta - \alpha p_k)}. \tag{2}$$

The parameter vector (β, α) can be estimated by maximum likelihood at the consumer level if individual choice data are available. With aggregate market-share data, the model is operationalised by inverting (2) as in Section 4 below.

Three assumptions are doing the work, and each carries a substantive empirical limitation. First, errors are i.i.d. across products: this generates the *independence of irrelevant alternatives* (IIA) restriction, under which the relative odds s_j/s_k depend only on j and k, not on what other products are in the choice set. This produces the textbook "red bus / blue bus" problem: introducing a second blue bus to a choice set with a car, a red bus, and a blue bus should mostly cannibalise the existing blue bus, but pure logit predicts proportional cannibalisation across all three options. In our cross-elasticities (Section 4.3) this manifests as identical off-diagonal entries within each column of the elasticity matrix, a strong restriction at odds with most product-differentiated industries. Second, there is no unobserved product quality: any structural error term ξ_j that shifts utility but is correlated with price will bias $\hat{\alpha}$ toward zero, because high-price products will (in equilibrium) also be high-ξ products and the regression cannot tell the two apart. Third, the model has no consumer heterogeneity in tastes: substitution patterns are governed entirely by mean utilities, so two consumers with very different demographics are assumed to make identical probabilistic choices. Random-coefficient models (BLP, Section 4 bonus discussion) relax the second and third restrictions; nested logit relaxes the first.

4 Berry (1994) Inversion

To accommodate unobserved product quality, augment (1) with a product-specific quality shock ξ_j:

$$u_{ij} = \underbrace{x_j^\top \beta - \alpha p_j + \xi_j}_{\equiv \delta_j} + \varepsilon_{ij}. \tag{3}$$

The mean utility δ_j now bundles observed characteristics, price, and the unobservable. Under EV1 errors and the outside normalisation $\delta_0 = 0$, taking the ratio of the inside-good share to the outside-good share yields

$$\frac{s_j}{s_0} = \exp(\delta_j) \quad \Longrightarrow \quad \log s_j - \log s_0 = x_j^\top \beta - \alpha p_j + \xi_j. \tag{4}$$

This is the Berry inversion: a single linear equation in observable shares (left-hand side) and observable characteristics, price, and the unobservable (right-hand side). The model is appealing because it admits unobserved heterogeneity in product appeal while retaining a tractable closed form. The drawback is that price p_j is now mechanically endogenous: any firm setting prices in equilibrium responds to its own ξ_j, so $\mathbb{E}[p_j\xi_j] \neq 0$. This breaks OLS but is identified with appropriate instruments.

4.1 OLS estimates

Estimating (4) by OLS treats price as exogenous. Table 3 column (1) reports estimates with $y_j = \log s_j - \log s_0$ as the dependent variable and $(p_j, ch_{2,j}, ch_{3,j}, ch_{4,j})$ as regressors. The implied price coefficient $\hat{\alpha}_{\text{OLS}} = -2.659$ has the expected sign and is precisely estimated. All three characteristics enter positively and significantly, with ch_2 the most valued and ch_4 the least. Goodness of fit is high ($R^2 = 0.937$), but in a model with simulation-clean instruments, fit alone does not certify the price coefficient.

The first-order issue is endogeneity. If unobserved quality ξ_j is positively correlated with price, as it is whenever firms with better products charge more, then OLS attributes part of the quality effect to price, biasing $\hat{\alpha}$ *upward in magnitude* (more negative than the truth). The IV estimate in column (2) is in fact slightly less negative than OLS, consistent with this story. A second issue is that OLS standard errors do not account for the equilibrium origin of the data: shares, prices, and characteristics are jointly determined, so heteroskedasticity-robust inference would be more honest, though point estimates are unaffected.

4.2 2SLS estimates

For the 2SLS estimation I use the two cost shifters as excluded instruments for price. The exclusion restriction is the standard cost-shifter argument: a variable that shifts marginal cost but does not enter consumer utility must satisfy $\mathbb{E}[\xi_j z_j] = 0$ while remaining correlated with p_j through the firm's pricing FOC. The relevance condition is testable; the exclusion is an economic argument that the cost shifter operates only on the supply side. Two cost shifters give one over-identifying restriction, which I use to test the exclusion restriction via Sargan.

Table 3 column (2) reports the 2SLS estimates. The price coefficient drops in magnitude to $\hat{\alpha}_{\text{IV}} = -2.552$, a reduction of 4% relative to OLS. The standard error rises by a factor of 3.4, the standard cost of trading exogeneity for efficiency. The first-stage F-statistic is 12.07, exceeding the conventional Stock–Yogo threshold of 10 for a single endogenous regressor with two instruments, so weak-instrument concerns are mild rather than acute. The Sargan over-identification statistic is 0.127 ($p = 0.72$); the data give no evidence that the instruments violate the exclusion restriction. The Wu–Hausman test of endogeneity does not reject ($p = 0.46$), but with the IV point estimate moving in the theoretically expected direction and the Sargan test passing, I report and use $\hat{\alpha}_{\text{IV}}$ throughout the policy exercises, consistent recovery of the structural price coefficient is what counterfactual analysis requires.

4.3 Market-10 derivatives and elasticities

Under the logit demand system in (2), the matrix of price derivatives within a single market admits a closed form:

$$\frac{\partial s_j}{\partial p_k} = \begin{cases} \hat{\alpha}_{\text{IV}}\, s_j(1 - s_j), & j = k \\ -\hat{\alpha}_{\text{IV}}\, s_j s_k, & j \neq k. \end{cases} \tag{5}$$

Note the matrix is symmetric: $\partial s_j / \partial p_k = \partial s_k / \partial p_j$. Elasticities are $\eta_{jk} = (\partial s_j / \partial p_k) \cdot (p_k / s_j)$. With $\hat{\alpha}_{\text{IV}} = -2.552$ and the observed shares for market 10, Tables 4 and 5 report the 11×11 derivative and elasticity matrices respectively.

The two matrices encode the same demand information measured in different units. The derivative matrix D is in share-per-price units; the elasticity matrix η is unit-free. The elasticities make the IIA restriction visible: every off-diagonal entry in any given column is identical. Substitution to product k is proportional across all rivals $j \neq k$, regardless of how similar each rival is to k. This is the empirical fingerprint of the type-I extreme value error structure. Whether the restriction is innocuous depends on the application: in a well-designed snack-food

Table 3: Berry inversion estimates. Dependent variable: $\log s_j - \log s_0$. Standard errors in parentheses. Diagnostics report the first-stage F-statistic for weak instruments, the Wu–Hausman test for endogeneity, and the Sargan test for over-identification.

	(1) OLS	(2) 2SLS
Intercept	0.849	0.749
	(0.265)	(0.300)
Price	-2.659	-2.552
	(0.046)	(0.153)
ch_2	1.632	1.526
	(0.129)	(0.195)
ch_3	1.158	1.082
	(0.129)	(0.166)
ch_4	0.508	0.505
	(0.075)	(0.076)
Instruments	–	2 cost shifters
First-stage F	–	12.07 ($p < 0.001$)
Wu–Hausman	–	0.55 ($p = 0.46$)
Sargan (overid.)	–	0.13 ($p = 0.72$)
R^2	0.937	0.935
N	247	247

experiment with substantial product differentiation, IIA is likely violated; in a market where the relevant differentiation is captured well by the included characteristics, the restriction may be benign. The own-price elasticities range from -6.04 (product 6, the largest market-10 product with $s = 0.147$) to -12.63 (product 10, with $s = 0.0022$): consistent with the logit feature that small-share products have steeper demand curves at the observed price.

5 Multi-Product Bertrand Markups

5.1 Supply-side model and derivation

I assume the supply side is multi-product Bertrand-Nash in posted prices. Each firm f chooses prices for its product portfolio $\mathcal{F}_f$ to maximise total profit:

$$\Pi_f(\mathbf{p}) = \sum_{k \in \mathcal{F}_f} (p_k - c_k)\, s_k(\mathbf{p}), \tag{6}$$

where c_k is the constant marginal cost of product k. The first-order condition for product $j \in \mathcal{F}_f$ is

$$\frac{\partial \Pi_f}{\partial p_j} = s_j(\mathbf{p}) + \sum_{k \in \mathcal{F}_f} (p_k - c_k)\, \frac{\partial s_k(\mathbf{p})}{\partial p_j} = 0. \tag{7}$$

Table 4: Market-10 derivative matrix $\partial s_j/\partial p_k$. Diagonal entries are own-price derivatives (negative); off-diagonals are cross-price derivatives (positive: products are substitutes). Rows are j, columns are k.

	k=1	k=2	k=3	k=4	k=5	k=6	k=7	k=8	k=9	k=10	k=11
j=1	**−0.043**	0.006	0.001	0.001	0.002	0.003	0.000	0.000	0.005	0.000	0.000
j=2	0.006	**−0.321**	0.005	0.004	0.020	0.024	0.002	0.001	0.039	0.001	0.001
j=3	0.001	0.005	**−0.036**	0.000	0.002	0.002	0.000	0.000	0.004	0.000	0.000
j=4	0.001	0.004	0.000	**−0.027**	0.001	0.002	0.000	0.000	0.003	0.000	0.000
j=5	0.002	0.020	0.002	0.001	**−0.127**	0.009	0.001	0.000	0.014	0.001	0.000
j=6	0.003	0.024	0.002	0.002	0.009	**−0.154**	0.001	0.000	0.017	0.001	0.000
j=7	0.000	0.002	0.000	0.000	0.001	0.001	**−0.016**	0.000	0.002	0.000	0.000
j=8	0.000	0.001	0.000	0.000	0.000	0.000	0.000	**−0.006**	0.001	0.000	0.000
j=9	0.005	0.039	0.004	0.003	0.014	0.017	0.002	0.001	**−0.237**	0.001	0.001
j=10	0.000	0.001	0.000	0.000	0.001	0.001	0.000	0.000	0.001	**−0.010**	0.000
j=11	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.000	**−0.005**

Table 5: Market-10 elasticity matrix $\eta_{jk} = (\partial s_j/\partial p_k)(p_k/s_j)$. Diagonals are own-price elasticities; off-diagonals are cross-price elasticities. Note the IIA pattern: every off-diagonal entry within column k is identical, substitution to product k is proportional across all rivals.

	k=1	k=2	k=3	k=4	k=5	k=6	k=7	k=8	k=9	k=10	k=11
j=1	**−7.53**	1.18	0.15	0.08	0.49	0.42	0.08	0.02	0.75	0.05	0.02
j=2	0.13	**−6.80**	0.15	0.08	0.49	0.42	0.08	0.02	0.75	0.05	0.02
j=3	0.13	1.18	**−10.37**	0.08	0.49	0.42	0.08	0.02	0.75	0.05	0.02
j=4	0.13	1.18	0.15	**−7.12**	0.49	0.42	0.08	0.02	0.75	0.05	0.02
j=5	0.13	1.18	0.15	0.08	**−8.81**	0.42	0.08	0.02	0.75	0.05	0.02
j=6	0.13	1.18	0.15	0.08	0.49	**−6.04**	0.08	0.02	0.75	0.05	0.02
j=7	0.13	1.18	0.15	0.08	0.49	0.42	**−11.47**	0.02	0.75	0.05	0.02
j=8	0.13	1.18	0.15	0.08	0.49	0.42	0.08	**−10.49**	0.75	0.05	0.02
j=9	0.13	1.18	0.15	0.08	0.49	0.42	0.08	0.02	**−6.50**	0.05	0.02
j=10	0.13	1.18	0.15	0.08	0.49	0.42	0.08	0.02	0.75	**−12.63**	0.02
j=11	0.13	1.18	0.15	0.08	0.49	0.42	0.08	0.02	0.75	0.05	**−9.36**

Stack the J_m FOCs from a single market into a matrix system. Define

$$\Omega_{jk} = \mathbf{1}\{f(j) = f(k)\}, \qquad \text{(ownership matrix)} \tag{8}$$

$$D_{jk} = \frac{\partial s_j}{\partial p_k}, \qquad \text{(derivative matrix)} \tag{9}$$

both $J_m \times J_m$. Under logit demand, D is symmetric (cf. (5)), so the FOC system collapses to

$$\mathbf{s} + (\Omega \circ D)\,\boldsymbol{\mu} = 0, \tag{10}$$

where $\circ$ is the Hadamard (element-wise) product and $\boldsymbol{\mu} = \mathbf{p} - \mathbf{c}$ is the vector of markups. Solving,

$$\boxed{\boldsymbol{\mu} = -(\Omega \circ D)^{-1}\,\mathbf{s}.} \tag{11}$$

The marginal cost is recovered residually as $\mathbf{c} = \mathbf{p} - \boldsymbol{\mu}$ and the Lerner index is $\boldsymbol{\mu}/\mathbf{p}$. The formula has two intuitive features. First, when a firm offers a single product, $\Omega \circ D$ is a diagonal matrix on that firm's row/column, and (11) reduces to the textbook single-product markup $\mu_j = -1/[\hat{\alpha}(1 - s_j)]$. Second, when a firm offers multiple products, the off-diagonal terms of

$\Omega \circ D$ pull markups upward: the firm internalises that raising the price of one of its products diverts demand to its other products, not just to rivals. Cross-market derivatives are zero by data construction, so the system can be solved one market at a time.

A property of pure logit that is worth flagging is that *multi-product firms set the same markup on every product they offer in a given market*. To see this, divide FOC (7) by s_j and substitute the logit derivatives:

$$1 + \hat{\alpha}\,\mu_j - \hat{\alpha} \sum_{k \in \mathcal{F}_f} \mu_k s_k = 0 \quad \Longrightarrow \quad \mu_j = \sum_{k \in \mathcal{F}_f} \mu_k s_k - \tfrac{1}{\hat{\alpha}}.$$

The right-hand side does not depend on j, so all products in firm f in market m carry the same markup. This is a known restriction of pure logit and is one of the motivations for moving to richer demand systems in counterfactual work.

5.2 Recovered markups

Table 6 reports the overall distribution of markups, implied marginal costs, and Lerner indices across all 247 products. Markups are tightly clustered between \$0.39 and \$0.55, with a median of \$0.41. The implied marginal cost distribution is wider, min \$0.76, max \$5.75, which is sensible: products differ much more in cost than in market power. The median Lerner index is 0.121, with mean 0.132 and a heavy right tail reaching 0.41 for the highest-power product. Lerner ≥ 0.41 corresponds to a multi-product firm with a large within-market portfolio.

Table 6: Distribution of markups, implied marginal costs, and Lerner indices across $J = 247$ products.

	Min	Q1	Median	Mean	Q3	Max
Markup	0.39	0.40	0.41	0.43	0.45	0.55
Marginal cost	0.76	2.45	3.12	3.07	3.65	5.75
Lerner index	0.07	0.10	0.12	0.13	0.15	0.41

Table 7 reports the product-level breakdown for market 10. The within-firm equal-markup feature appears clearly. Products 237 and 240 are both owned by firm 1; both carry markup \$0.4030. Products 245 and 247 are both owned by firm 3; both carry markup \$0.4381. Products 239 and 246 are both owned by firm 11; both carry markup \$0.3992. The cross-firm differences are governed by the firms' total within-market shares: firm 3 (combined market-10 share 0.106) sets the highest markup, while firm 11 (combined share 0.018) sets the lowest among multi-product firms.

6 MERGER SIMULATION: FIRMS 1 AND 10

6.1 Welfare arguments for and against the merger

The proposed merger has both pro-competitive and anti-competitive arguments.

Arguments for. The clearest case is the failing-firm defense. The assignment frames firm 1 as "struggling" and the merger as a bailout: under the failing-firm doctrine in U.S. horizontal merger guidelines, a deal that prevents an inevitable exit can be welfare-improving even with substantial overlap, because the counterfactual is not the status quo but the loss of all of

Table 7: Market-10 product-level markups, marginal costs, and Lerner indices. Within-firm products carry identical markups under pure logit, as derived in the text.

Product	Firm	Price	Share	Markup	MC	Lerner
237	1	3.00	0.017	0.40	2.60	0.13
238	6	3.12	0.147	0.46	2.67	0.15
239	11	4.12	0.014	0.40	3.73	0.10
240	1	2.82	0.011	0.40	2.42	0.14
241	8	3.64	0.053	0.41	3.23	0.11
242	5	2.53	0.064	0.42	2.11	0.17
243	2	4.52	0.007	0.39	4.13	0.09
244	10	4.12	0.002	0.39	3.73	0.10
245	3	2.84	0.104	0.44	2.40	0.15
246	11	4.97	0.004	0.40	4.57	0.08
247	3	3.67	0.002	0.44	3.24	0.12

the failing firm's products. A merger also enables the combined entity to consolidate fixed costs, rationalise duplicative back-office functions, and spread R&D investment over a larger base; if these efficiencies are real and pass through to prices, consumers can be net better off. Finally, scale can support investment in product quality or innovation that neither firm could afford alone. The empirical question for an antitrust authority is whether such efficiencies are merger-specific (achievable only through the merger) and verifiable, rather than vague claims of "synergy."

Arguments against. The standard concern is unilateral price effects: merging firms internalise substitution between their formerly separate product portfolios and raise prices on overlapping products. The simulation in Section 6.2 quantifies this channel directly. Beyond unilateral effects, market concentration can facilitate tacit coordination among the remaining firms, especially if the merger reduces the number of significant rivals and the post-merger market is relatively transparent. Reduced product variety is a third concern: a combined firm may rationalise its catalogue, and consumers who valued the discontinued products lose. Finally, a merger that creates a dominant firm can foreclose entry or raise rivals' costs through downstream or upstream leverage, even if the immediate price effects appear modest.

The right framework for weighing these considerations is the merger-guidelines balance: efficiencies must be cognisable, merger-specific, and pass-through to consumers; harms include direct price effects, coordination risks, and dynamic effects on innovation and entry. The structural simulation below speaks directly to the unilateral-effects channel.

6.2 Counterfactual simulation

Solution method. The merger holds demand parameters and marginal costs fixed and modifies only the ownership structure. The post-merger ownership matrix is

$$\Omega_{jk}^{\text{post}} = \begin{cases} 1, & \text{if } f(j) = f(k), \text{ or both } f(j), f(k) \in \{1, 10\}, \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

The price-free part of mean utility, $\eta_j \equiv x_j^\top \beta + \xi_j = \log s_j^{\text{obs}} - \log s_0^{\text{obs}} - \hat{\alpha}\, p_j^{\text{obs}}$, is invariant to the merger; so is marginal cost c_j. Post-merger equilibrium prices solve the fixed-point system

$$\mathbf{p}^{\text{post}} = \mathbf{c} + \boldsymbol{\mu}(\mathbf{p}^{\text{post}}), \qquad \boldsymbol{\mu}(\mathbf{p}) = -[\Omega^{\text{post}} \circ D(\mathbf{p})]^{-1}\, \mathbf{s}(\mathbf{p}), \tag{13}$$

where $\mathbf{s}(\mathbf{p})$ and $D(\mathbf{p})$ are evaluated at $\delta_j(\mathbf{p}) = \eta_j + \hat{\alpha}\, p_j$ each iteration. I solve (13) market by market via damped iteration: $\mathbf{p}^{(t+1)} = (1-\lambda)\mathbf{p}^{(t)} + \lambda\, [\mathbf{c} + \boldsymbol{\mu}(\mathbf{p}^{(t)})]$ with damping $\lambda = 0.5$, initialised at observed prices, and stopped at $\|\mathbf{p}^{(t+1)} - \mathbf{p}^{(t)}\|_\infty < 10^{-10}$. Convergence is fast and stable: 26–30 iterations across the ten markets. Pre-merger share recovery achieves machine precision (max absolute error $\approx 10^{-16}$), confirming that the structural primitives are recovered correctly.

Price and markup effects. Table 8 reports averages by ownership group. Merging-firm products see an average price increase of \$0.034 (1.0% of the pre-merger price), with a median increase of \$0.019. Outside-firm products barely move: the average price change is \$0.0003 (0.01%), driven by the small share of customers who substitute toward outside firms after the merging firms raise prices. Average post-merger market shares fall slightly for the merging entity (−0.15 percentage points per product) and rise for outside firms (essentially zero on a per-product basis but positive in aggregate). The pattern is the textbook unilateral-effects signature: the merging entity raises prices, the resulting share losses are partially captured by outside firms, and the total inside-share for the market falls (consumers exit toward the outside option).

Table 8: Post-merger pricing and quantity effects, averaged within ownership group across all 10 markets. "Merging" refers to firms 1 and 10; "outside" refers to firms 2–9 and 11.

Group	Products	Mean Δp	Median Δp	Mean $\%\Delta p$	Mean $\Delta\mu$	Mean Δs
Merging	52	0.0340	0.0190	1.011%	0.0340	−0.0015
Outside	195	0.0003	0.0002	0.010%	0.0003	0.0002
All	247	0.0074	0.0003	0.221%	0.0074	−0.0002

Consumer welfare. Under the logit, the dollar-denominated per-consumer compensating variation is

$$\Delta\text{CV}_m = -\frac{1}{\hat{\alpha}}\left[\text{IV}_m^{\text{post}} - \text{IV}_m^{\text{pre}}\right], \qquad \text{IV}_m(\mathbf{p}) = \log\Big(1 + \sum_{k\in m} e^{\eta_k + \hat{\alpha}\, p_k}\Big), \tag{14}$$

where IV_m is the inclusive value (log-sum-exp) of market m. With $\hat{\alpha} < 0$ and price increases lowering inclusive value, $\Delta\text{CV}_m < 0$: consumers are unambiguously worse off in every market that experiences any price effect. Without market-size data I cannot scale to total dollar harm, so I report per-consumer effects.

Table 9 reports market-level CV losses. Per-consumer harm is concentrated where the merging firms' combined market position is strongest. Market 9 shows the largest per-consumer loss (−\$0.0168), driven by firm 10's combined 20% inside share in that market (largest single-firm position firm 10 holds anywhere). Markets 6 and 7 are next, where firm 10 holds 17% and 14% shares respectively. The smallest losses are in markets where firms 1 and 10 are minor players: market 5 (−\$0.000002) and market 10 (−\$0.00005), where the combined merging position is below 4%.

Table 9: Per-consumer compensating variation by market. Negative values indicate consumer harm. Mean across all ten markets is $-\$0.0042$ per consumer.

Market	ΔCV per consumer	IV$^{\text{pre}}$	IV$^{\text{post}}$
1	-0.0001	0.907	0.907
2	-0.0029	0.830	0.823
3	-0.0063	1.527	1.512
4	-0.0010	0.757	0.754
5	-0.0000	0.932	0.932
6	-0.0078	0.687	0.667
7	-0.0068	1.759	1.742
8	-0.0004	1.404	1.403
9	-0.0168	0.541	0.498
10	-0.0001	0.552	0.552

Magnitude in context. The mean across markets is $-\$0.0042$ per consumer, modest as a share of the average pre-merger price ($\sim$ 0.1%). Three forces drive the small magnitude. First, the market is competitive ex ante: with 11 firms and substantial inter-firm substitution captured by the IIA cross-elasticities, even a two-firm merger leaves nine independent rivals. Second, firms 1 and 10 have only modest combined within-market shares in most markets, their combined position exceeds 20% only in market 9. Third, pure logit implies relatively flat substitution patterns: with no random coefficients, products are not especially close substitutes for one another, so the merger's diversion effect is dampened. A BLP-style demand system with rich consumer heterogeneity would likely produce larger and more localised price effects, particularly in markets where the merging firms' products are perceived as close substitutes by particular consumer segments.

Market-10 detail. Table 10 reports product-level results for market 10. Firm 1's products 237 and 240 rise from markup \$0.4030 to \$0.4038, a modest 0.03% price increase. Firm 10's product 244 rises from \$0.3927 to \$0.4038 (+0.27%). All three merging products converge to identical post-merger markups, exactly as the within-firm logit property predicts: post-merger they are now in the same firm, so they receive the same markup. Outside products move only at the fourth decimal place. Market 10 is the smallest-effect market in the dataset; it is also the one with the smallest combined position (firms 1 and 10 together hold 2.98% inside share), confirming the relationship between market position and merger impact.

7 REFLECTIONS

7.1 Application: hospital merger in a regional metropolitan market

I would apply this methodology to evaluate a proposed merger between two regional hospital systems serving the same metropolitan area, for example, the kind of Chicago-area transaction in which a large academic medical center proposes to acquire a community-hospital network that overlaps in two or three counties. Hospital mergers are a natural setting because the relevant "products" are hospitals (or hospital service lines), patient choice is observed at the discharge level, and the antitrust authorities have an established framework for evaluating them. The

Table 10: Market-10 product-level effects of the simulated merger between firms 1 and 10. Products from firms 1 and 10 are flagged in the third column.

Product	Firm	Merged?	p^{pre}	p^{post}	$\%\Delta p$	μ^{pre}	μ^{post}	$\Delta\mu$
237	1	yes	3.0032	3.0041	0.029%	0.4030	0.4038	0.0009
238	6	no	3.1247	3.1247	0.000%	0.4596	0.4596	0.0000
239	11	no	4.1242	4.1242	0.000%	0.3992	0.3992	0.0000
240	1	yes	2.8202	2.8211	0.031%	0.4030	0.4038	0.0009
241	8	no	3.6430	3.6430	0.000%	0.4136	0.4136	0.0000
242	5	no	2.5307	2.5307	0.000%	0.4188	0.4188	0.0000
243	2	no	4.5244	4.5244	0.000%	0.3944	0.3944	0.0000
244	10	yes	4.1188	4.1299	0.270%	0.3927	0.4038	0.0111
245	3	no	2.8410	2.8410	0.000%	0.4381	0.4381	0.0000
246	11	no	4.9676	4.9676	0.000%	0.3992	0.3992	0.0000
247	3	no	3.6740	3.6740	0.000%	0.4381	0.4381	0.0000

implementation maps onto the methodology used in this problem set as follows.

Data. Discharge-level claims data identify the patient (with diagnosis, age, insurance status, and zip code), the hospital chosen, and the contracted price paid by the insurer. Hospital characteristics include service offerings, teaching status, technology indicators (catheterisation lab, MRI, robotic surgery), bed count, quality measures (HCAHPS scores, mortality rates), and ownership type. The market is defined geographically, typically using a Hospital Service Area or commuting-zone definition, and the outside option is the patient's choice not to use any in-system hospital (out-of-network, out-of-area, or non-utilisation). Cost shifters include input cost variation across hospitals (wages, supplies indices), regional capacity constraints, and exposure to differential reimbursement schedules.

Demand. Patient choice is modelled via a discrete choice framework parallel to Section 4, with one important departure: patients face zero or near-zero out-of-pocket prices at the point of service for in-network hospitals, so "price" in the textbook sense is not the right shifter for utility. Instead, demand is driven by hospital characteristics, network coverage (whether the patient's insurance contracts with the hospital), travel distance, and patient-hospital match characteristics (prior visits, physician affiliations). A BLP-style random-coefficients specification is essential here because patient heterogeneity is first-order: a young patient with a maternity stay is a fundamentally different consumer than a 70-year-old cardiac patient, and IIA across these segments is implausible.

Supply. The supply side requires the most substantial deviation from PS1's setup, and it is the topic of Section 7.2. Briefly: hospitals do not post prices and consumers do not respond to them. Prices are set in bilateral negotiations between each hospital and each managed care organisation, with patients flowing to the hospital chosen by the network/quality/distance bundle. The right model is Nash-in-Nash bargaining à la Horstmann and Markusen (1989); Crawford and Yurukoglu (2012); Gowrisankaran et al. (2015), in which each hospital–insurer pair bargains over price, taking other negotiated prices as given.

Counterfactual. The merger simulation modifies the bargaining structure rather than just the ownership matrix. When two hospitals merge, they bargain jointly with each insurer; the disagreement value for the insurer becomes worse, because exclusion now means losing access to both hospitals at once. This raises the negotiated price even in markets where unilateral pricing power is small, because the leverage operates through the insurer's outside option (loss of network breadth). The simulation reports per-insurer price changes and patient-welfare effects, with the latter requiring assumptions about insurer pass-through to premiums.

Why this is the right test case. Hospital mergers generate observable post-merger data (closed transactions, ex-post price effects), there is a substantial empirical literature against which to benchmark estimates (Gaynor, Ho), and the regulatory framework (FTC and DOJ joint guidelines, applied to hospitals through the Hospital Merger Guidelines) is concrete. The methodology in PS1 maps directly onto the analytical machinery, with the bargaining adjustment being the principal modification.

7.2 Critique from the opposing side

If I were retained by the merging hospitals to oppose a structural-merger-simulation analysis presented by the FTC, I would attack the methodology on six grounds.

1. The supply model is wrong. The single most serious critique is that hospital prices are negotiated, not posted. The PS1 supply model assumes Bertrand-Nash in posted prices. Applied to hospitals it produces a markup formula (11) that is mechanically misspecified: hospitals do not post a price and observe patient choices on a downward-sloping demand curve. Hospitals bargain bilaterally with insurers, taking patient routing as given. The right model is the Horn–Wolinsky (Horn and Wolinsky, 1988) Nash-in-Nash framework formalised for hospital–insurer pairs by Gowrisankaran et al. (2015). Nash-in-Nash delivers a fundamentally different markup expression, in which prices depend on bargaining weights and the disagreement values of both parties. A Bertrand-Nash markup applied to hospital data is not just biased, it is the wrong object. This would be the lead point of attack in litigation, and it cannot be patched by tweaking the demand side.

2. IIA in the demand system understates close substitution. Pure logit, and even most aggregate BLP specifications, embed restrictions that systematically understate how close two hospitals are as substitutes for each other. In hospital choice, two hospitals serving the same neighbourhood with overlapping service lines and shared physicians are far closer substitutes than the general logit cross-elasticity allows. The merger simulation's price effects are mechanically a function of these cross-elasticities; understating them understates the unilateral-effects harm. The right answer is patient-level random coefficients with a rich set of distance, service-line, and demographic interactions, but even then the structural restriction may bind.

3. Market definition is contested. The choice of geographic market (HSA, MSA, county, commuting zone) materially affects the diversion ratio between the merging hospitals and therefore the predicted price effect. The opposing side can present sensitivity analyses across alternative market definitions and argue that the FTC's choice is conservative or selective. The PS1 setup avoids this by giving us markets exogenously; in practice, market definition is the most-litigated step.

4. The failing-firm defense. If one of the merging hospitals is in genuine financial distress, a frequent claim in community-hospital acquisitions, the relevant counterfactual is not the status quo but the hospital's exit, in which case some patients lose access entirely and others migrate to the more-distant remaining hospitals. The structural simulation as performed compares post-merger to pre-merger; it does not compare post-merger to exit. The opposing side would argue that the correct welfare comparison is to the no-acquisition counterfactual, which may itself involve reduced access and worse health outcomes.

5. Efficiencies and quality. Mergers in healthcare often involve cost-side or quality-side efficiencies that the demand system cannot capture: scale in medical-records systems, broader physician recruiting, capital investment in technology, and centralisation of complex specialty services. If the merger reduces marginal cost or raises quality (a positive shock to the ξ_j component of utility), then the post-merger equilibrium may exhibit lower prices or higher welfare than the simulation suggests. The simulation holds both demand and cost primitives fixed across the merger, which is exactly the assumption efficiencies-defenders dispute.

6. External validity to the legal standard. Merger litigation requires showing that the merger "substantially lessens competition" under Section 7 of the Clayton Act. A structural simulation produces point estimates with model-imposed standard errors; the evidentiary standard in court asks whether the harm is substantial, given the entire range of plausible model specifications. A defending team would present a battery of specifications, different demand systems, different supply assumptions, different market definitions, showing that price effects can be small or near-zero under reasonable alternatives. The structural model's strength (precise quantitative predictions) is also its litigation weakness: it offers many surfaces on which to challenge specification choices.

The honest meta-point is that all six critiques apply with full force to PS1's data only because we are extending the methodology to a domain it was not built for. Within the PS1 framework, posted prices, simulated data, and clean instruments, the simulation is valid and informative. The lesson for ex-ante merger work is that the structural model's domain of applicability is narrower than its computational tractability suggests, and applied to the wrong industry it produces precise, internally consistent, and substantively wrong answers.

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